Informatica® Intelligent Data Management Cloud

April 2026

Data Services Repository

Informatica Intelligent Data Management Cloud Data Services Repository
April 2026

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Preface

Use *Data Services Repository* to learn how to manage a repository of data services that process industry-standard and custom messages.

Informatica Resources

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The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Data services repository

The data services repository is a central repository of data services that process industry-standard messages. If your organization exchanges customized messages with your partners, you can upload custom data services to the repository to process the custom messages.

The data services repository includes more than 50,000 out-of-the-box data services, including the following data services and versions:

- ACORD AL3 versions Fall 2009, Fall 2011, Spring 2012
- BAI versions BAI2 and Lockbox
- EDIFACT versions 90A to 09A
- EDI-X12 versions 3010 to 7010
- HIPAA version 5010A1
- HIX version 5010
- HL7 all versions
- NACHA version 2022
- NCPDP versions 5.1 and D.0
- SWIFT MT, MX-ISO20022 versions SR21, SR22, SR23, SR24

For example, use the HIPAA data services to exchange HIPAA messages with your healthcare partners.

Additionally, the repository includes HL7 to FHIR mapper and validation services.

If your organization or your partners customize messages by adding or removing message elements, create custom data services to process the custom messages. For example, if a partner uses customized Z-segments in the messages, create custom services to process the non-standard messages.

You use the data services in a Data Services transformation in a mapping. For more information about the Data Services transformation, see *Transformations* in the Data Integration help.

Data service names and types

The name of a data service indicates the industry standard, the version of the standard, the message type, and the service usage type, in the following syntax:

```
<industry standard>_<version>_<message type>_<usage type>
```

The usage type can be one of the following types:

- Parser. Use a parser service to receive messages from your partners.

- Serializer. Use a serializer service to send messages to your partners.
- Restricted serializer. Use a restricted serializer service to send messages to your partners, including validation messages. Applies to HIPAA messages.

For example, to build HIPAA message 270, version 5010A1, use the following service:

HIPAA_5010A1_270_Parser

Data service customization

You use custom data services to process customized messages.

To create a custom data service, perform the following tasks:

1. In Administrator, download an out-of-the-box data service for the standard form of the same message or for a similar message and unzip the download file to a folder with a unique name.

   Note: You can't edit an out-of-the-box service. You must download it and create a new service.
2. Edit the files that you downloaded to align with the customized message structure.
3. In Administrator, create a custom data service and associate the folder of the customized service files with it.

Downloading a data service

You can download both custom and out-of-the-box data services from the data services repository to your local machine. After you download an out-of-the-box service, you can use the service as a template to base a custom service on.

1. On the **Data Services Repository** page, select the row that contains the service, click the Download icon, and then save the ZIP file to your local machine.
2. Unzip the file to a folder with a unique name.

   Folder names are case sensitive.

Creating a custom data service

When you create a custom data service, you upload a data service folder or files that you created based on an out-of-the-box service.

1. On the **Data Services Repository** page, click **New Custom Data Service**.

   The **New Custom Data Service** dialog box opens.
2. Browse to select the folder that contains the service files.

   Note: You can't upload files with the following file extensions: `app, bat, bin, cmd, com, dll, exe, inf, jar, msi, net, osx, out, py, run, script, sh, shortcut, so, vbs, vbscript, wsh, x86, x86_64, php`. Files can't contain double extensions.

3. Optionally, add a description to the service. The description can't contain the following special characters: < , >.
4. Select the industry standard, version, message type, operation, and environment.

   Note: HIPAA uses the following message subtypes for message type 837:

   - Version 222: Message type 837P
   - Version 223: Message type 837I
   - Version 224: Message type 837D

   If the version contains BHT and 11 for message type 278, the message subtype is 278_RES or 278_REQ otherwise.

   If the version contains TH for message type 277, the message subtype is 277CA.
5. Click **Save**.

   The data service appears on the **Data Services Repository** page. It takes about five minutes to deploy a data service.
6. To view the custom data service, filter the data services by **Type: Custom**.

Editing or deleting a custom data service

You can edit a custom data service to associate a different service folder with the service. The folder must exist before you edit the service. You can't edit or delete an out-of-the-box data service.

1. To edit a custom data service, on the **Data Services Repository** page, select the row that contains the service and then click the Edit icon.
2. To associate a different data service folder or files with the service, browse to select the folder that contains the data service files.
3. Optionally, select a different industry standard, version, message type, operation, or environment.

   Note: HIPAA uses the following message subtypes for message type 837:

   - Version 222: Message type 837P
   - Version 223: Message type 837I
   - Version 224: Message type 837D

   If the version contains BHT and 11 for message type 278, the message subtype is 278_RES or 278_REQ otherwise.

   If the version contains TH for message type 277, the message subtype is 277CA.
4. Click **Save**.

To delete a custom data service on the **Data Services Repository** page, select the row that contains the service and then click **Delete**.

Saving and resetting default filters

Save default filters to automatically filter the data services each time you visit the **Data Services Repository** page. Reset the filters to the default filters to revert any changes to the filters.

1. Filter the data services.
2. To save the filters as the default filters, click **Save Default Filters.**
3. To revert to the default filters, click **Reset to Default Filters**.

The default filters apply on the **Data Services Repository** page and in the Data Services transformation in Data Integration.

CHAPTER 2

Data service reference

Data services include parsers and serializers that can process industry-standard messages. They can parse messages and write the data to another format like XML, or serialize data and convert it to a message.

Use the data service reference to find descriptions of the out-of-the-box data services available in the data services repository.

ACORD AL3

The ACORD AL3 data services implement the AL3 messaging standard used in the insurance industry. AL3 is maintained by the Association for Cooperative Operations Research and Development (ACORD), a nonprofit insurance association. AL3 messages handle the transmission of insurance information such as policies and claims.

ACORD AL3 message structure

ACORD has a positional format that uses question marks as placeholders for missing data.

The following example shows an ACORD message that handles watercraft insurance:

```
1MHG176   IBM716ESTEDL      ??????????????????231225               AFW
50270000010406111100051           70
2004061111000512TRG212   70         3P PBOAT FMG
0001040611        040511 PCH
20040611200405112TCG135
2TAG212   00003278                    P06260822
WTRP  PRODUCER
2ACI200   Stephen F. Newman, Inc.
1600 Delaware Avenue, Suite 999 Seneca NY1499916837168563387
5BIS172   B10001             P        Ester W.         Powell
????????????????????????????????00003278
??????????????????????
9BIS168   B10001             3672 Almond drive
Hopatka NY14888    7166090166
5ISI183   B200015BISB10001          ??   ?????????? ???????????????
5SNG117   B200015BISB10001    01DECJane W. Fieldstone
???1??????????????????
5BPI285   F100015BISB10001    P06260822               ??????????
FISH 040511050511
```

ACORD AL3 message in XML format

The ACORD AL3 parser data services can convert ACORD messages into XML format. The following example shows an ACORD message in XML format:

```
<AL3_301_WaterCraft>
  <Message_Header_Group>
    <Header_00>1MHG176</Header_00>
    <Message_Address_Origination_01>IBM716ESTEDL</Message_Address_Origination_01>
```

```
    <Message_Address_Destination_02>??????????????</Message_Address_Destination_02>
    <Contract_Number_03>231225</Contract_Number_03>
    <Password_User_04 />
    <System_Type_Code_05>AFW</System_Type_Code_05>
    <Interface_Software_Revision_Level_06>5027</Interface_Software_Revision_Level_06>
    <Message_Sequence_Number_07>000001</Message_Sequence_Number_07>
    <Message_Transmission_Date_Time_08>0406111100051</Message_Transmission_Date_Time_08>
    <Count_Unit_Code_09 />
    <Special_Handling_10 />
    <Message_Standard_Revision_Level_11>70</Message_Standard_Revision_Level_11>
    <Transmission_Status_Flag_12 />
    <Message_Response_Code_13 />
    <Error_Code_14 />
    <Network_Reference_Code_15 />
    <Network_Reserved_For_Future_Use_16 />
    <Message_Transmission_Date_Time_17>200406111100051</
Message_Transmission_Date_Time_17>
  </Message_Header_Group>
```

BAI

The BAI data services implement specifications developed by the BAI financial services industry organization. They include Cash Management Balance Reporting Specifications Version 2 (BAI2) and Lockbox Communications Standards for Banks (BAI Lockbox).

BAI2 message structure

BAI2 messages have a record format. Each record ends with a line break and contains comma-delimited fields.

The first two digits of each record indicate the record type. For example, notice the record types in the following BAI2 message:

```
01,122099999,123456789,040621,0200,1,65,,2/
02,031001234,122099999,1,040620,2359,,2/
03,0123456789,,010,+4350000,,,040,2830000,,/
88,072,1020000,,,074,500000,,/
16,115,450000,S,100000,200000,150000,,,/
16,115,10000000,S,5000000,4000000,1000000/
88,AX13612,B096132,AMALGAMATED CORP. LOCKBOX
88,DEPOSIT-MISC. RECEIVABLES
49,9150000,4/
03,9876543210,,010,-500000,,,100,1000000,,,400,2000000,,,190/
88,500000,,,110,1000000,,,072,500000,,,074,500000,,,040/
88,-1500000,,/
16,115,500000,S,,200000,300000,,,LOCK BOX NO.68751
49,4000000,5/
98,13150000,2,11/
02,053003456,122099999,1,040620,2359,,2/
03,4589761203,,010,10000000,,,040,5000000,,,074,4000000,,/
...
```

The `01` record is a file header that contains information such as the sender and receiver IDs. The `02` records designate account groups. The other records contain data for specific bank transactions. The `88` records are continuation lines.

BAI2 message in XML format

The BAI2 parser data services can convert BAI2 messages into XML format. The following example shows a BAI2 message in XML format:

```
<BAI>
  <Delimiters>
```

```
    <RecordDelimiter>/</RecordDelimiter>
    <FieldDelimiter>,</FieldDelimiter>
  </Delimiters>
  <FileHeader_01>
    <SenderIdentification>072000096</SenderIdentification>
    <ReceiverIdentification>072000096</ReceiverIdentification>
    <FileCreationDate>060322</FileCreationDate>
    <FileCreationTime>0433</FileCreationTime>
    <FileIdentificationNumber>1</FileIdentificationNumber>
    <PhysicalRecordLength>80</PhysicalRecordLength>
    <VersionNumber>2</VersionNumber>
  </FileHeader_01>
  <LoopGroups>
    <GroupHeader_02>
      <UltimateReceiverIdentification>000000000</UltimateReceiverIdentification>
      <OriginatorIdentification>072000096</OriginatorIdentification>
      <GroupStatus>1</GroupStatus>
      <AsOfDate>060321</AsOfDate>
      <AsOfTime>0433</AsOfTime>
      <CurrencyCode />
      <AsOfDateModifier>2</AsOfDateModifier>
    </GroupHeader_02>
    <LoopAccounts>
      <AccountIdentifier_03>
        <CustomerAccountNumber>1000000001</CustomerAccountNumber>
        <CurrencyCode />
        <TypeCode>015</TypeCode>
        <Amount>+0000016668216</Amount>
        <FundsType />
        <TypeCode>040</TypeCode>
        <Amount>+0000016668216</Amount>
```

BAI Lockbox message structure

BAI Lockbox messages have a positional structure, such as in the following example:

```
1004121092548012100024806112105O5
241210925480121000248000000000040008000801
500100102017450611214121092548012100024B
600100100003779691110006146345669700 071324    MAYDAY
4001001601034145204       0000016750
4001001602034145409       0000009500
```

BAI Lockbox message in XML format

The BAI Lockbox parser data services can convert BAI Lockbox messages into XML format. The following example shows a BAI Lockbox message in XML format:

```
<BAI_Lockbox>
  <ImmediateAddressHeader>
    <PriorityCode>00</PriorityCode>
    <ImmediateDestination>4121092548</ImmediateDestination>
    <OriginCode>0121000248</OriginCode>
    <DepositDate>061121</DepositDate>
    <TransmissionTime>0505</TransmissionTime>
  </ImmediateAddressHeader>
  <ServiceRecord>
    <UltimateDestination>4121092548</UltimateDestination>
    <OriginCode>0121000248</OriginCode>
    <ReferenceCode>0000000000</ReferenceCode>
    <ServiceType>400</ServiceType>
    <RecordSize>80</RecordSize>
    <BlockingFactor>0080</BlockingFactor>
    <FormatCode>1</FormatCode>
  </ServiceRecord>
  <Lockbox>
    <DetailHeader>
      <BatchNumber>001</BatchNumber>
      <ItemNumber>001</ItemNumber>
      <LockboxCode>0201745</LockboxCode>
      <DepositDate>061121</DepositDate>
```

```
    <UltimateDestination>4121092548</UltimateDestination>
    <OriginCode>0121000248</OriginCode>
  </DetailHeader>
  <Batch>
    <DetailRecord>
      <BatchNumber>1</BatchNumber>
      <ItemNumber>1</ItemNumber>
      <RemittanceAmount>377969</RemittanceAmount>
      <RemitterIdentification>11100061</RemitterIdentification>
```

EDIFACT

The EDIFACT data services implement the EDIFACT international electronic data interchange standard maintained by the United Nations.

EDIFACT message structure

An EDIFACT message contains segments and fields that are separated by a hierarchy of delimiters, such as in the following example:

```
UNA:+,? '
UNB+UNOA:2+BANESTO+TELEFONICA DE ESPAÑA S.A.+050321:1204+0030200503211'
UNH+20050321120412+BANSTA:098:96A:UN'
BGM+XZ8+20050321120412+9'
DTM+137:20050321:102'
RFF+ACW:20050321113331'
DTM+171:20050321:102'
LIN+1'
SEQ+YF2+1'
GIS+1'
DTM+140:20050321:102'
MOA+9:0,00'
CNT+2:1'
CNT+X27:1'
AUT+INTERV49         107835'
UNT+14+20050321120412'
UNZ+1+0030200503211'
```

EDIFACT message in XML format

The EDIFACT parser data services can convert EDIFACT messages into XML format. The following example shows an EDIFACT message in XML format:

```
<interchange>
  <delimiters>
    <field_separator>+</field_separator>
    <segment_separator>'</segment_separator>
    <composite_separator>:</composite_separator>
    <escape_character>?</escape_character>
  </Delimiters>
  <UNA>
    <R01>:</R01>
    <R02>+</R02>
    <R03>,</R03>
    <R04>?</R04>
    <R05 />
    <R06>'</R06>
  </UNA>
  <UNB>
    <R01>
      <R01>UNOA</R01>
      <R02>2</R02>
    </R01>
    <R02>
```

```
    <R01>BANESTO</R01>
  </R02>
```

EDI-X12

The EDI-X12 data services contain components that convert messages between X12 and XML and generate validation error reports and acknowledgements. X12 is a standard for electronic data interchange (EDI) between trading partners over data networks. X12 is developed and maintained by the Accredited Standards Committee (ASC).

X12 message structure

An X12 message contains the following layers:

- Interchange. The outer layer that wraps the entire X12 message.
- Functional group. The middle layer that wraps one or more transaction sets.
- Transaction set. The inner layer that contains the data.

Each transaction set contains a header, one or more data segments, and a trailer. A transaction set might contain data such as a purchase order, an invoice, or a statement of an account.

The following sample message shows an X12 Vessel Content Details transaction set:

```
ST*109*0001~B4*15*0*2*19960515*1424*Statu*Equi*Equipment*D*XXXX*
Location Identifier*A*3~N9*01*Reference Identification*
Free-form Description*19960515*1424*01*01^Reference Identification^01^
Reference Identification^01^Reference Identification~Q2*X*XX*19960515*
19960515*19960515*10045*21392*A*Flight/Voy*01*Reference Identification*
B*Vessel Name*10691*B*E~V9*AAD*Event*19960515*1424*City Name*XX*XX*001*
XXXXXX*25946*Tr*Free-Form Message*01*25878*XXXXXX*717*437*272*2457*
12935~R4*1*A*Location Identifier*Port Name*XX*Terminal Name*Pier*XX~DTM*
001*19960515*1424*01*CC*Date Time Period~V9*AAD*Event*19960515*1424*
City Name*XX*XX*001*XXXXXX*4685*Tr*Free-Form Message*01*13647*XXXXXX*813*
605*52*20035*12104~N9*01*Reference Identification*Free-form Description*
19960515*1424*01*01^Reference Identification^01^Reference Identification^
01^Reference Identification~SG*2*001*XX*19960515*1424*01~SE*11*0001~
```

Segments are delimited by tildes (~) and data elements are delimited by asterisks (*).

EDI-X12 message in XML format

The EDI-X12 parser data services can convert X12 messages into XML format. The following example shows an X12 message in XML format:

```
<interchange>
  <delimiters>
    <field_separator>*</field_separator>
    <segment_separator>~</segment_separator>
    <composite_separator>^</composite_separator>
    <escape_character>?</escape_character>
  </Delimiters>
  <ISA>
    <R01>00</R01>
    <R02 />
    <R03>00</R03>
    <R04 />
    <R05>01</R05>
    <R06>003897733</R06>
    <R07>ZZ</R07>
    <R08>SLRGATEWAY</R08>
    <R09>050202</R09>
    <R10>1338</R10>
    <R11>U</R11>
```

```
    <R12>00401</R12>
    <R13>000000708</R13>
```

FHIR

The FHIR (Fast Healthcare Interoperability Resources) data services implement the FHIR standard used in the healthcare industry. FHIR is developed and maintained by HL7 International, a nonprofit organization dedicated to establishing standards for the exchange of electronic health information. FHIR handles healthcare information such as patient records, treatments, and billing details.

FHIR message structure

FHIR messages are packaged in bundles that can contain resources with details about patient information, observations, or treatment details. FHIR messages use XML format, such as in the following example:

```
<?xml version="1.0" encoding="UTF-8"?>

<Patient xmlns="http://hl7.org/fhir">
  <id value="example"/>
  <!--    MRN assigned by ACME healthcare on 6-May 2001    -->
  <identifier>
    <use value="usual"/>
    <type>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
        <code value="MR"/>
      </coding>
    </type>
    <system value="urn:oid:1.2.36.146.595.217.0.1"/>
    <value value="12345"/>
    <period>
      <start value="2001-05-06"/>
    </period>
    <assigner>
      <display value="Acme Healthcare"/>
    </assigner>
  </identifier>
  <active value="true"/>
  <!--    Peter James Chalmers, but called "Jim"    -->
  <name>
    <use value="official"/>
    <family value="Chalmers"/>
    <given value="Peter"/>
    <given value="James"/>
  </name>
  <name>
    <use value="usual"/>
    <given value="Jim"/>
  </name>
  <name>
    <!--   Maiden names apply for anyone whose name changes as a result of marriage -
irrespective
     of gender   -->
    <use value="maiden"/>
    <family value="Windsor"/>
    <given value="Peter"/>
    <given value="James"/>
    <period>
      <end value="2002"/>
    </period>
  </name>
  <!--    home communication details aren't known    -->
  <telecom>
    <use value="home"/>
```

```
</telecom>
<telecom>
  <system value="phone"/>
  <value value="(03) 5555 6473"/>
  <use value="work"/>
  <rank value="1"/>
</telecom>
<telecom>
  <system value="phone"/>
  <value value="(03) 3410 5613"/>
  <use value="mobile"/>
  <rank value="2"/>
</telecom>
<telecom>
  <system value="phone"/>
  <value value="(03) 5555 8834"/>
  <use value="old"/>
  <period>
    <end value="2014"/>
  </period>
</telecom>
<!--    use FHIR code system for male / female    -->
<gender value="male"/>
<birthDate value="1974-12-25">
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
    <valueDateTime value="1974-12-25T14:35:45-05:00"/>
  </extension>
</birthDate>
<deceasedBoolean value="false"/>
<address>
  <use value="home"/>
  <type value="both"/>
  <text value="534 Erewhon St PeasantVille, Rainbow, Vic  3999"/>
  <line value="534 Erewhon St"/>
  <city value="PleasantVille"/>
  <district value="Rainbow"/>
  <state value="Vic"/>
  <postalCode value="3999"/>
  <period>
    <start value="1974-12-25"/>
  </period>
</address>
<contact>
  <relationship>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v2-0131"/>
      <code value="N"/>
    </coding>
  </relationship>
  <name>
    <family value="du Marché">
      <!--    the "du" part is a family name prefix (VV in iso 21090)    -->
      <extension url="http://hl7.org/fhir/StructureDefinition/humanname-own-prefix">
        <valueString value="VV"/>
      </extension>
    </family>
    <given value="Bénédicte"/>
  </name>
  <additionalName>
    <use value="nickname"/>
    <given value="Béné"/>
  </additionalName>
  <telecom>
    <system value="phone"/>
    <value value="+33 (237) 998327"/>
  </telecom>
  <address>
    <use value="home"/>
    <type value="both"/>
    <line value="534 Erewhon St"/>
    <city value="PleasantVille"/>
```

```
      <district value="Rainbow"/>
      <state value="Vic"/>
      <postalCode value="3999"/>
      <period>
        <start value="1974-12-25"/>
      </period>
    </address>
    <additionalAddress>
      <use value="work"/>
      <line value="123 Smart St"/>
      <city value="PleasantVille"/>
      <state value="Vic"/>
      <postalCode value="3999"/>
    </additionalAddress>
    <gender value="female"/>
    <period>
      <!--    The contact relationship started in 2012    -->
      <start value="2012"/>
    </period>
  </contact>
  <managingOrganization>
    <reference value="Organization/1"/>
  </managingOrganization>
</Patient>
```

You can use the FHIR validation service from the data services repository to validate the XML structure of a FHIR message before writing it to a FHIR server.

HIPAA

The HIPAA data services implement the HIPAA messaging standard for administrative and financial health-care transactions. HIPAA is based on the X12 standard.

The data services repository includes HIPAA parser and serializer data services, as well as data services that validate input messages and generate acknowledgements. These data services implement the complete set of validation and acknowledgment types defined in the HIPAA standard up through version 5010A1.

HIPAA message structure

The following example shows a HIPAA message for a Benefit Enrollment and Maintenance transaction that an insurance sponsor uses to provide benefit enrollment information to payers:

```
ISA*00**00**ZZ*EDI0021*ZZ*54828*032403*1253*U*00405*000000905*1*T*:
GS*BE*EDI0021*54828*20030324*1253*5*X*004050X125~
ST*834*26608~
BGN*00*26608*20021226*1200*ES*2~
N1*P5*UES CORP*FI*111222333~
N1*IN*UNITED HEALTHCARE*FI*777888999~
INS*Y*18*021*28*A***FT~
REF*0F*851027461~
REF*1L*199301~
DTP*356*D8*20021219~
NM1*IL*1*CANOE*NANCY*A*MRS.*I*99*99999999~
PER*IP**HP*4104038790*WP*4104035500*EX*5684~
N3*9999 ROSE VILLAGE*FIRST FLOOR~
N4*FLOWERS*ND*999995*CY*WHEELING~
DMG*D8*19740801*F*M~
HD*021**PPO*PLAN COVERAGE DESCRIPTION*ESP~
DTP*348*D8*20021219~
SE*16*26608~
ST*834*26608~
BGN*00*26608*20021226*1200*ES*2~
N1*P5*UES CORP*FI*111222333~
```

```
N1*IN*UNITED HEALTHCARE*FI*777888999~
INS*N*01*021*28*A***FT~
REF*0F*948047150~
REF*1L*199301~
```

HIPAA message in XML format

The HIPAA parser data services can convert HIPAA messages into XML format. The following example shows a HIPAA message in XML format:

```
<interchange xmlns="http://www.informatica.com/B2B/HIPAA/4010" xmlns:H834="http://
www.informatica.com/B2B/HIPAA/4010/834" >
  <delimiters>
    <field_separator>*</field_separator>
    <composite_separator>:</composite_separator>
    <segment_separator>!</segment_separator>
  </Delimiters>
  <ISA>
    <R01>00</R01>
    <R02 />
    <R03>00</R03>
    <R04 />
    <R05>ZZ</R05>
    <R06>101018</R06>
    <R07>ZZ</R07>
    <R08>107039</R08>
    <R09>04101</R09>
    <R10>1334</R10>
```

HIX

The HIX data services implement the ASCX12 healthcare industry standard. They use the Health Insurance Exchange protocol to exchange information for government-regulated and standardized health care plans in the United States.

The HIX protocol is based on the HIPAA standard, and data services can process HIPAA messages that are relevant to the ASCX12 standard with Validation Level 1 and 2.

Note: Don't use the HIX data services to validate HIPAA or EDI-X12 messages. The HIX data services contain unique validators to process ASCX12 messages.

HIX message structure

A HIX message is a string that contains fields delimited by special characters. Each field has an identifier followed by data.

The following example shows a sample Related Payments message type (820):

```
ISA*00*..........*01*SECRET....*ZZ*SUBMITTERS.ID..*ZZ*RECEIVERS.ID...*030101*1253*^*00501
*000000905*1*T*:~
GS*RA*SENDER CODE*RECEIVERCODE*19991231*0802*1*X*005010X306~
ST*820*0001*005010X306~
BPR*C*310*C*ACH*CCP*******01*199999999*DA*98765*20121103~
TRN*3*78905~
N1*PE*BATA INSURANCE CO.*FI*012222222~
ENT*1~
NM1*IL*1*DOE*JOHN****C1*777222~
REF*IG*555666~
RMR*ZZ*1**30~
ENT*2~
NM1*IL*1*FIRSTONE*EMILY****C1*777333~
REF*IG*555777~
RMR*ZZ*1**45~
```

```
ENT*3~
NM1*EY*1*MIDDLEONE*JULIE****C1*777444~
REF*IG*544477~
RMR*ZZ*1**50~
DTM*582****RD8*20120901-20121031~
ENT*4~
NM1*EY*1*NEWONE*KELLY****EI*777111~
RMR*ZZ*2**450~
SE*27*0001~
GE*1*1~
IEA*1*000000905~
```

HIX message in XML format

The HIX parser data services can convert HIX messages into XML format. The following example shows a HIX message in XML format:

```
  <?xml version="1.0" encoding="windows-1252" ?>
- <X5010:interchange xmlns:X5010="http://www.informatica.com/B2B/HIX/5010"
xmlns:X5010_820="http://www.informatica.com/B2B/HIX/5010/820">
    - <delimiters>
      <field_separator>*</field_separator>
      <segment_separator segment_separator_suffix="">~</segment_separator>
      <composite_separator>:</composite_separator>
      <repeat_separator />
      </delimiters>
- <X5010:ISA>
   <R01>00</R01>
   <R02>..........</R02>
   <R03>01</R03>
   <R04>SECRET....</R04>
   <R05>ZZ</R05>
   <R06>SUBMITTERS.ID..</R06>
   <R07>ZZ</R07>
   <R08>RECEIVERS.ID...</R08>
   <R09>030101</R09>
   <R10>1253</R10>
   <R11>^</R11>
   <R12>00501</R12>
   <R13>000000905</R13>
   <R14>1</R14>
   <R15>T</R15>
   <R16>:</R16>
  </X5010:ISA>
- <X5010:GS>
   <R01>RA</R01>
   <R02>SENDER CODE</R02>
   <R03>RECEIVERCODE</R03>
   <R04>19991231</R04>
   <R05>0802</R05>
   <R06>1</R06>
   <R07>X</R07>
   <R08>005010X306</R08>
  </X5010:GS>
- <X5010:GE>
   <R01>1</R01>
   <R02>1</R02>
   </X5010:GE>
- <X5010:IEA>
   <R01>1</R01>
   <R02>000000905</R02>
   </X5010:IEA>
  </X5010:interchange>
```

HL7

The HL7 data services implement the Health Level Seven version 2.x messaging standard used in the health services industry. The HL7 standard is used world-wide in hospital and medical information systems.

HL7 message structure

An HL7 message contains segments that are separated by line breaks. Each segment has a three-character label, such as `MSH` (message header) or `PID` (patient identification). Each segment contains a predefined hierarchy of fields and sub-fields. The fields are delimited by the characters immediately following the `MSH` designator, typically `|^~\&`. The message type is specified by a field in the `MSH` segment.

The following example shows a message of type `ADT` and subtype `A01`, which is an `Admit a Patient` message:

```
MSH|^~\&|ADT1|MCM|FINGER|MCM|198808181126|SECURITY|ADT^A01|MSG00001|P|2.3.1
EVN|A01|198808181123
PID|1||PATID1234^5^M11^ADT1^MR^MCM~123456789^^^USSSA^SS||
SMITH^WILLIAM^A^III||19610615|M||C|1200 N ELM STREET^^JERUSALEM^TN^99999?
1020|GL|(999)999?1212|(999)999?3333||S||PATID12345001^2^M10^ADT1^AN^A|
123456789|987654^NC
NK1|1|SMITH^OREGANO^K|WI^WIFE||||NK^NEXT OF KIN
PV1|1|I|2000^2012^01||||004777^CASTRO^FRANK^J.|||SUR||||ADM|A0
```

The patient's name, `Smith^William^A`, follows the `PID` label by five `|` delimiters. The last and first names, `Smith` and `William`, are separated by a `^` delimiter.

For readability, the sample message contains extra line breaks in addition to the line breaks that are in the original message.

HL7 message in XML format

The HL7 parser data services can convert HL7 messages into HL7 version 2.x XML schemas. The following example shows an HL7 message in XML format:

```
<HL7:ADT_A01 xmlns:HL7="http://www.informatica.com/B2B/HL7">
  <HL7:MSH>
    <MSH.1>|</MSH.1>
    <MSH.2>^~\&</MSH.2>
    <MSH.3>
      <HD.1>ADT1</HD.1>
    </MSH.3>
    <MSH.4>
      <HD.1>MCM</HD.1>
    </MSH.4>
    <MSH.5>
      <HD.1>LABADT</HD.1>
    </MSH.5>
    <MSH.6>
      <HD.1>MCM</HD.1>
    </MSH.6>
    <MSH.7>
      <TS.1>198808181126</TS.1>
    </MSH.7>
    <MSH.8>SECURITY</MSH.8>
    <MSH.9>
      <MSG.1>ADT</MSG.1>
      <MSG.2>A01</MSG.2>
    </MSH.9>
    <MSH.10>MSG00001</MSH.10>
```

NACHA

The NACHA data services implement the Automated Clearing House standard maintained by the NACHA Electronic Payments Association and used by financial institutions.

NACHA message structure

A NACHA message contains positional records. The following example shows a sample NACHA message:

```
101 091000022 1130006090208240516A094101US BANK CORP PAY SYS   CHASE OF
TX
5200BEST WIDGETS USA                    1234567890CTXPAYMENT
020828240111300060000066
602152545851FDHHDFHDFHDFHFHFH1254565236              1254ADRFVFRTG5467HYJ
1125456523658745
602152545851FDHHDFHDFHDFHFHFH1254565236              1254ADRFVFRTG5467HYJ
1125456523658745
```

NACHA message in XML format

The NACHA parser data services can convert NACHA messages into XML format. The following example shows a NACHA message in XML format:

```
<File>
  <FileHeaderRecord>
    <PriorityCode>1</PriorityCode>
    <ImmediateDestination>91000022</ImmediateDestination>
    <ImmediateOrigin>113000609</ImmediateOrigin>
    <FileCreationDate>20824</FileCreationDate>
    <FileCreationTime>516</FileCreationTime>
    <FieldIdModifier>A</FieldIdModifier>
    <RecordSize>094</RecordSize>
    <BlockingFactor>10</BlockingFactor>
    <FormatCode>1</FormatCode>
    <ImmediateDestinationName>US BANK CORP PAY SYS</ImmediateDestinationName>
    <ImmediateOriginName>CHASE OF TX</ImmediateOriginName>
    <ReferenceCode />
  </FileHeaderRecord>
  <Batch>
    <BatchHeaderRecord>
      <ServiceClassCode>200</ServiceClassCode>
      <Name>BEST WIDGETS USA</Name>
      <DiscretionaryData />
      <Identification>1234567890</Identification>
      <StandardEntryClassCode>CTX</StandardEntryClassCode>
      <EntryDescription>PAYMENT</EntryDescription>
      <DescriptiveDate />
      <EffectiveEntryDate>20828</EffectiveEntryDate>
      <SettlementDateJulian>240</SettlementDateJulian>
      <OriginatorStatusCode>1</OriginatorStatusCode>
      <OriginatingDfiIdentification>11300060</OriginatingDfiIdentification>
      <BatchNumber>666</BatchNumber>
    </BatchHeaderRecord>
    <Record>
      <CTX>
        <CTXEntryDetailRecord>
          <TransactionCode>2</TransactionCode>
          <ReceivingDfiIdentification>15254585</ReceivingDfiIdentification>
          <CheckDigit>1</CheckDigit>
          <DfiAccountNumber>FDEHHFDHDFHDFHFHFH</DfiAccountNumber>
```

NCPDP

The NCPDP data services implement the messaging protocols used in the pharmacy services sector of the healthcare industry. The National Council for Prescription Drug Programs (NCPDP) develops these protocols to improve the transmission of pharmacy-related data, including prescription transactions, medication history, and eligibility verification. Insurance companies use NCPDP transactions to process pharmacy drug claims.

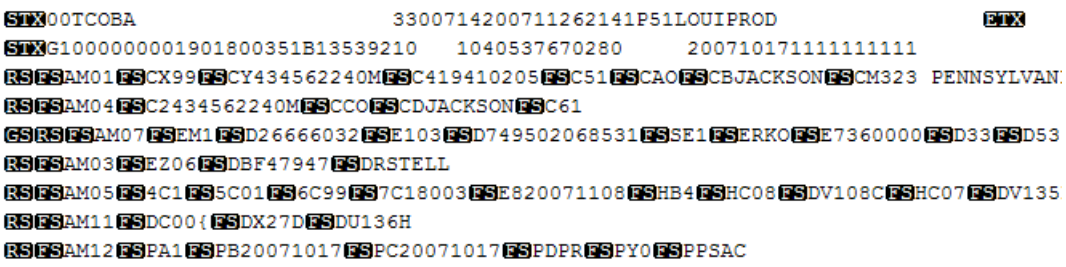
The data services repository includes the following NCPDP standards for communicating pharmacy claims:

- NCPDP Batch Transaction Standard versions 1.1 and 1.2 to transmit multiple claims in batch format.
- NCPDP Telecommunication Standard Format versions 5.1 and D.0 to transmit individual claims from the point-of-sale to the insurance carrier.

NCPDP message structure

NCPDP messages have a positional layout that contains both fixed and variable-length fields. The segments and fields are delimited by control characters such as STX (ASCII 0x02), ETX (0x03), FS (0x1C), GS (0x1D), and RS (0x1E).

The following image shows part of an NCPDP billing message:



NCPDP message in XML format

The NCPDP parser data services can convert NCPDP messages into XML format. The following example shows an NCPDP message in XML format:

```
<File>
  <Batch>
    <BatchHeaderRecord>
      <TransmissionType>Transaction</TransmissionType>
      <SenderID>COBA</SenderID>
      <BatchNumber>3300714</BatchNumber>
      <CreationDate>20071126</CreationDate>
      <CreationTime>2141</CreationTime>
      <FileType>Production</FileType>
      <VersionReleaseNumber>51</VersionReleaseNumber>
      <ReceiverID>LOUIPROD</ReceiverID>
    </BatchHeaderRecord>
    <DetailDataRecord>
      <TransactionReferenceNumber>0000000019</TransactionReferenceNumber>
      <TransmissionTransactionType="Billing">
        <Header>
          <BinNumber>18003</BinNumber>
          <VersionReleaesNumber>51</VersionReleaesNumber>
```

SWIFT

The SWIFT data services implement the SWIFT messaging standard for the financial industry, maintained by the Society for Worldwide Interbank Financial Telecommunication. The standard defines messages for purposes such as electronic funds transfer and check processing.

The data services repository includes the following SWIFT data services:

- SWIFT MT for text-based messages
- SWIFT MX for XML-based messages

SWIFT MT message structure

The following example shows a Request for Transfer message for a SWIFT MT transaction:

```
{1:F01DEUTDEFFEDIX0317000005}{2:I101ABCDEFGHIJKAU1020}{4:
:20:Sender Ref Dom01
:21R:Customer Ref
:28D:1/1
:50L:Instructing Party
:50H:/00190020974010046074
DB SPAIN
:30:030120
:21:REFF21-1
:32B:EUR1,00
:57D:/ES999999999
Madrid
:59:/64930056505
DB Spain
Barcelona
:70:First payment11
:71A:OUR
:21:REFF21-2
:32B:EUR2,00
:57D:/E999999999
Madrid
:59:/64930056506
DB Spain
Barcelona
:70:Second payment2
:71A:OUR
```

SWIFT MT message in XML format

The SWIFT MT parser data services can convert SWIFT MT messages into XML format. The following example shows a SWIFT MT message in XML format:

```
<MT101>
  <BasicHeaderBlock>
    <ApplicationIdentifier>F</ApplicationIdentifier>
    <ServiceIdentifier>01</ServiceIdentifier>
    <LTAddress>
    </LTAddress>
    <SessionNumber></SessionNumber>
    <SequenceNumber></SequenceNumber>
  </BasicHeaderBlock>
  <ApplicationHeaderInputBlock>
    <Indicator>I</Indicator>
    <MessageType>101</MessageType>
    <DestinationAddress>
      <BankCode>ABCD</BankCode>
      <CountryCode>EF</CountryCode>
      <LocationCode>GH</LocationCode>
      <LogicalTerminalCode>1</LogicalTerminalCode>
      <BranchCode>JKA</BranchCode>
    </DestinationAddress>
    <MessagePriority>U</MessagePriority>
    <DeliveryMonitoring>1</DeliveryMonitoring>
```

```
  <ObsolescencePeriod>020</ObsolescencePeriod>
</ApplicationHeaderInputBlock>
```

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